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A SIMPLIFIED NUMERICAL BENCHMARK FOR POOL-TYPE SODIUM FAST REACTORS

Stefan Radman

Laboratory for Reactor Physics and Systems Behaviour Laboratory for Reactor Physics and Systems Behaviour

Ecole Polytechnique Fdrale de Lausanne
1005 Lausanne, Vaud, Switzerland
Email: stefan.radman@epfl.ch

Carlo Fiorina

Ecole Polytechnique Fdrale de Lausanne
1005 Lausanne, Vaud, Switzerland
Email: carlo.fiorina@epfl.ch

Konstantin Mikityuk

Laboratory for Reactor Physics and Systems Behaviour Laboratory for Reactor Physics and Systems Behaviour

Paul Scherrer Institut
5232 Villigen, Aargau, Switzerland
Email: konstantin.mikityuk@psi.ch

Andreas Pautz

Ecole Polytechnique Fdrale de Lausanne
1005 Lausanne, Vaud, Switzerland
Email: andreas.pautz@epfl.ch

ABSTRACT

The current work presents a simplified benchmark for a pool-type Sodium Fast Reactor based on a 2D (r-z) geometry for testing tightly-coupled spatial neutron kinetics, thermal-hydraulics and thermal-mechanics modeling. The new benchmark is motivated by development of the multi-physics OpenFOAM-based GeN-Foam code at the Laboratory for Reactor Physics and Systems Behaviour at the EPFL, Switzerland, and the FAST code system by the advanced nuclear systems group at the Paul Scherrer Institut, Switzerland. Aiming at the improvement of modeling and code-to-code comparison, the benchmark could prove useful for developers of tightly-coupled multi-physics simulation tools for reactor analysis. The benchmark specification and the solutions obtained with GeN-Foam will be presented and discussed in the paper.

INTRODUCTION

A simplified numerical benchmark methodology is proposed in the context of the development of tightly-coupled multi-

physics simulation tools for pool-type Sodium Fast Reactor (SFR) analysis. The goal of the present work is to: 1) outline and showcase a numerical benchmark methodology for the assessment of SFR multi-physics code performance through selected quantities of interest; 2) provide a tool to enable periodic regression tests of SFR multi-physics codes under development. Such a benchmark is envisioned to improve both the code development and code-to-code comparison processes. On the code development side, tight coupling in multi-physics codes may result in single-physics code updates having unforeseen consequences on the other physics. Thus, a tool for the assessment of code update effects (i.e. regression testing) through selected quantities is desirable. On the code-to-code comparison side, a consistent numerical benchmark methodology among institutions involved in code development would prove of great importance.

The present numerical benchmark methodology is set in the broader context of the ESRF-SMART project, dedicated to the safety assessment and computational tools improvement for the European SFR reactor concept. This work also stems from the internal activities on multi-physics code development of the Lab-

oratory for Reactor Physics and Systems Behaviour at the EPFL, Switzerland, and the FAST group at the Paul Scherrer Institut, Switzerland, specifically in relation to the OpenFOAM-based GeN-Foam code and the FAST code system.

The current preliminary numerical benchmark consists of: 1) a simplified 2-D pool-type SFR model; 2) a methodology to assess the performance of SFR-dedicated multi-physics code. In the present preliminary development stage, the multi-physics code used for the benchmark is the OpenFOAM-based GeN-Foam code, instrumental to demonstrating the methodology. Nonetheless, the rationale of the work is to provide a methodology that is straightforward to implement in any multi-physics platform of interest.

With regard to the present document, an overview of the coupled physics for SFR calculations is presented first. The computational environment of the employed multi-physics GeN-Foam code is then briefly presented. Subsequently, a description of the employed 2-D reactor geometry is provided. Stemming from such considerations, the benchmark methodology is proposed next. The results obtained by the application of the selected code to the benchmark are presented to further clarify the methodology. Conclusion regarding further developments constitute the final section of the present document.

MULTI-PHYSICS COUPLING OVERVIEW

Multi-physics codes for the modelling of nuclear systems rely on adequately coupled single-physics models. The involved physics consist in thermal-hydraulics, neutron kinetics and thermal-mechanics. A generic, non-code-specific overview of the different physics and their coupling is provided in Fig. 1, followed by a discussion on their interplay. The discussion is presented in terms of idealized single-physics solvers.

Thermal-hydraulics

The individual contributions from a fluid-dynamics solver and an energy solver can be split, as it provides a further layer of understanding of the overall physics coupling.

Fluid-dynamics Given a temperature field T over a computational domain Ω , a fluid-dynamics solver is dedicated to computing the resulting velocity and pressure fields. The temperature field affects the solver via the fluid molecular viscosity μ and fluid density ρ , which are generally temperature dependent quantities.

Energy Given velocity $\mathbf{u}$ and pressure p fields over a computational domain Ω , an energy solver is dedicated to computing the resulting temperature distribution in the system. The velocity field affects the solver via advective terms, while the

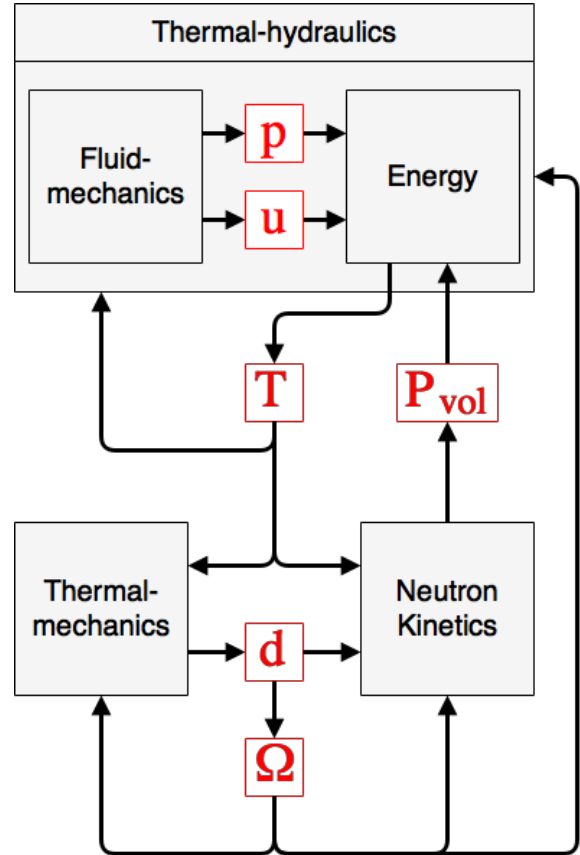


FIGURE 1. Overview of the multi-physics coupling scheme. The arrow directions indicate which variables are output by which physics, and which physics are affected by these variables. The variables that introduce the coupling are highlighted in red.

pressure field affects the solver via time derivative terms which can play a significant role in compressible flow scenarios. The source term for the energy equation consists in a volumetric heat source term, that is somehow reconstructed from a volumetric power field P_{vol} that can be provided by a neutron kinetics solver.

Thermal-mechanics

Given a temperature field within the fuel pins and core structures over a computational domain Ω , a thermal-mechanics solver is dedicated to computing the resulting displacement field $\mathbf{d}$. This displacement field is then employed to: 1) provide feedback to a neutron kinetics solver via the macroscopic cross sections (generically referred to as Σ), as the displacement field can capture fuel expansion phenomena and thus fuel density reduction; 2) displace the computational domain Ω .

Neutron Kinetics

Given a set of macroscopic cross sections, collectively referred to as Σ , over a computational domain Ω , a neutron kinetics solver is dedicated to computing the resulting neutron flux Φ . This forms the basis for the computation of the volumetric power field P_{vol} . Macroscopic cross sections are directly affected by the temperature field via the Doppler effect and indirectly by the fuel thermal expansion and subsequent fuel density change. The former effect is captured by parametrising the cross-sections with respect to temperature, while the latter can be captured by a further parametrisation with respect to the displacement field (e.g. axial fuel expansion, i.e. axial fuel displacement).

COMPUTATIONAL ENVIRONMENT

GeN-Foam is an OpenFOAM [1] based multi-physics code developed at the Laboratory for Reactor Physics and Systems Behaviour (LRS), EPFL. While a complete code description can be found in [2], [3], [4], the goal of the current section is to provide an overall introduction. This will be instrumental for further discussions regarding the benchmark model.

From the standpoint of thermal-hydraulics, GeN-Foam relies on a single-phase, porous medium based [5] coarse mesh treatment. A porous medium based coarse-mesh methodology allows for a simplified description of complex systems, while the interaction between the fluid phase and the geometrically unresolved sub-scale structures are captured via adequate closure models. In the fluid-dynamics solver, such an interaction is modelled through a volumetric drag force term, which relies on correlations for pressure drops. In the energy solver, the interaction is modelled through a volumetric heat source term. Such a term in turn relies on a 1-D heat conduction model for the computation of representative fuel and cladding temperatures within each mesh cell. The source term for the 1-D heat conduction model consists in the volumetric *fuel* power field P_{vol} . The aforementioned 1-D heat conduction model is included in the energy sub-solver.

From the standpoint of neutron kinetics, the solver can model: 1) transient scenarios through either a diffusion equation or an SP3 approach; 2) steady-state scenarios through an eigenvalue approach. The macroscopic cross sections are to be provided as an input, and can be parametrised with respect to: 1) axial fuel expansion; 2) cladding expansion; fuel temperature; 3) radial core expansion; 4) coolant density; 5) coolant temperature.

From the standpoint of thermal-mechanics, the purpose of the solver is to predict radial core expansion and axial fuel expansion. In particular, the axial fuel and cladding expansion is predicted by a separate 1-D displacement model within the solver. The temperature field for the computation of the displacement fields is provided by the thermal-hydraulics solver.

BENCHMARK MODEL

In this section, the simplified axial-symmetric 2-D ESFR model on which the benchmark is based is presented. The model consists of three different geometries and meshes, each bound to one of the specific physics. The OpenFOAM computational environment allows in fact to take advantage of standardized mesh-to-mesh mapping functions, which grant a great deal of flexibility in terms of individual mesh and geometry choices.

Thermal-hydraulics

The computational domain over which the fluid-dynamics and energy sub-solvers operate is reported in Fig. 2.

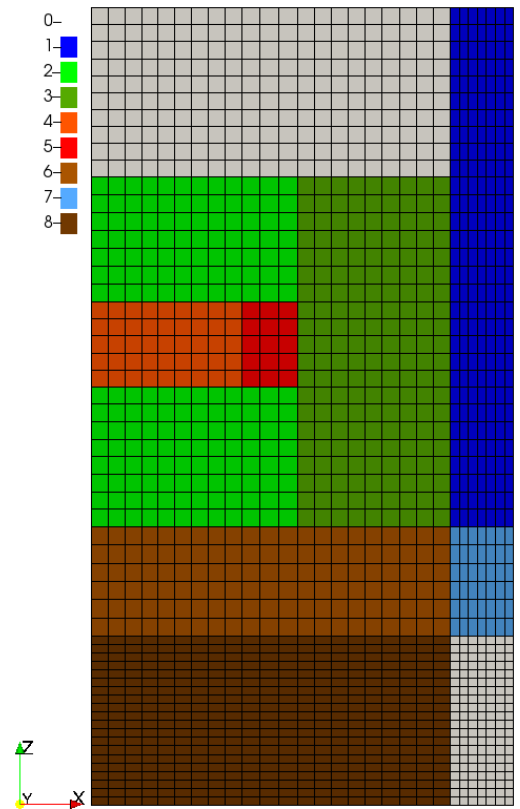


FIGURE 2. Computational domain for thermal-hydraulics. The symmetry axis is in the z direction, starting at the left-most system boundary. It is composed of the following regions: 0) liquid sodium, i.e. simplified hot and cold pools; 1) heat exchanger; 2) axial reflectors; 3) radial reflector; 4) inner core; 5) outer core; 6) diagrid; 7) pump; 8) strongback.

All of the regions are modelled as porous media, with the exception of the simplified sodium pools. In fact, one of the advantages provided by a coarse mesh approach is that it enables a uniform system description of both porous and non-porous regions.

In the pump, a uniform, downward volumetric momentum source term can be prescribed. In the heat exchanger, diagrid and radial reflector regions, values for the density, heat capacity and volumetric surface areas of the sub-scale structures are prescribed. This allows for the computation of representative sub-scale structure temperatures, which are mapped to the thermal-mechanics geometry for the computation of the displacement field. A heat transfer coefficient and heat sink temperature are prescribed in the heat exchanger for the modelling of a heat sink term. In the inner and outer core regions, the volumetric surface area of the geometrically unresolved fuel pins is prescribed. Coupled with the 1-D heat conduction model for the fuel pins, this provides a volumetric heat source term to the energy solver. Finally, a set of baffles separates the heat exchanger from the radial reflector, the pump from the diagrid and the radial reflector from both the axial reflectors and the outer core. A slip boundary condition is applied to all of the domain boundaries for the velocity field, as well as a zero gradient prescription for temperature and pressure.

Neutron Kinetics

The computational domain over which the neutron kinetics solver operates is reported in Fig. 3.

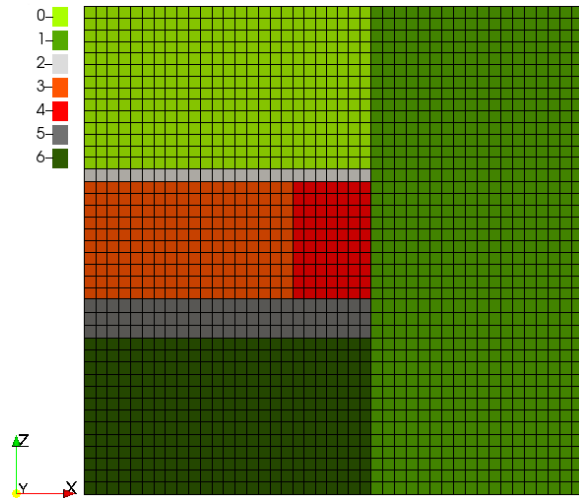


FIGURE 3. Computational domain for neutron kinetics. The symmetry axis is in the z direction, starting at the left-most system boundary. It is composed of the following regions: 0) upper axial reflector; 1) radial reflector; 2) upper gas plenum; 3) inner core; 4) outer core; 5) lower gas plenum; 6) lower axial reflector;

The lower and upper axial reflectors are modelled separately as they are characterized by different cross sections. Upper and lower gas plena are modelled as well. The neutron flux computed in the inner and outer core regions provides the basis for

the computation of the volumetric fuel power distribution. This is mapped onto the thermal-hydraulic mesh to provide a heat source term for the 1-D fuel pin model. The neutron flux is prescribed to vanish at the domain boundaries.

Thermal-mechanics

The computational domain over which the thermal-mechanics solver operates is reported in Fig. 3.

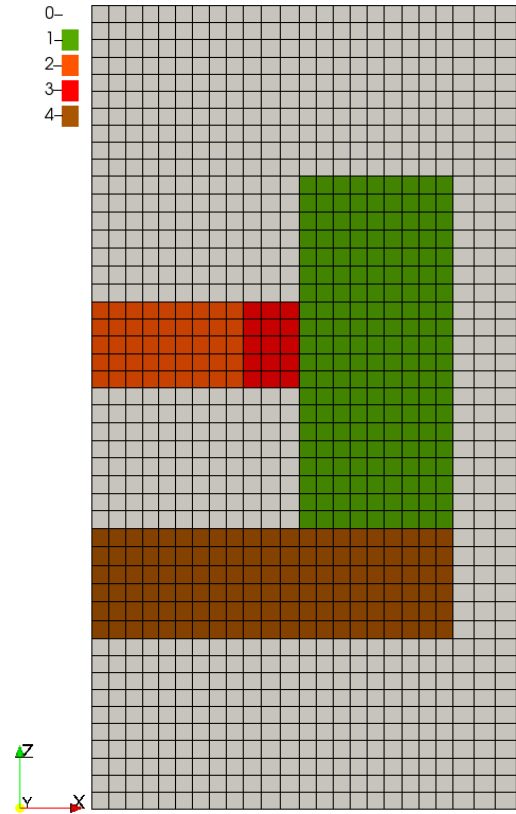


FIGURE 4. Computational domain for thermal-mechanics. The symmetry axis is in the z direction, starting at the left-most system boundary. It is composed of the following regions: 0) soft structures; 1) radial reflector; 2) inner core; 3) outer core; 4) diagrid;

All of the regions which are not relevant in shaping core radial expansion and axial fuel expansion are collectively described by a so-called "soft structure". It is thermo-mechanically described by a null thermal expansion coefficient and low elastic modulus, so to act as a region in which other elements can expand almost freely. Prescribed thermal expansion coefficients in the diagrid and radial reflector regions allow for the computation of the core radial expansion. In fact, for an SFR design perspective, the assemblies are supported by the diagrid, so that

a radial expansion of the diagrid directly results in a radial core expansions. The inner and outer core regions are characterized by a sub-scale model for the computation of the axial fuel expansion, with prescribed thermal expansion coefficients. The resulting displacement field in the domain is then used to displace the mesh. A null displacement field is prescribed at the upper and lower boundaries of the computational domain, while free expansion is allowed at the radial boundary.

BENCHMARK METHODOLOGY

Stemming from what was discussed up to this point, a numerical benchmark methodology is proposed. In the first phase, steady-state results are obtained by the single physics solvers with no coupling over prescribed input fields. Characteristic quantities specific to each physics, which will be introduced in the present section, are thus obtained. In the second phase, steady-state results are obtained by the solvers coupled in different combinations, and the characteristic quantities re-evaluated. This is done to isolate the coupling contributions. The last stage of this phase involves full coupling of all of the physics at steady-state.

In the remainder of the present section, the actual simulations stages are described in terms of the input fields to be provided and the output quantities to be obtained. As a final remark, the term "core" will be extensively used to refer to both the inner and outer core regions, thus exclusive of the axial and radial reflector regions.

Single-physics Simulations

I Fluid-dynamics The fluid-dynamics sub-solver is employed to compute the steady-state velocity field in the system. Input:

- uniform temperature field, constant in time;
- uniform pump momentum source, constant in time.

Output:

- total mass flow through core;
- maximum, minimum, average sodium velocity in the core;
- total pressure drop across the core (exclusive of hydrostatic component).

II Energy The energy sub-solver is employed to compute the steady-state temperature distribution in the system. These are inclusive of average core structure temperatures and fuel temperatures. Input:

- velocity field obtained in stage I, constant in time;
- uniform volumetric fuel power field in the core, constant in time.

Output:

- axial (radially integrated) temperature profile in the core;
- maximum, minimum and average coolant temperatures in the core;
- axial (radially integrated) density profile in the core;
- maximum, minimum and average coolant density in the core.

III Neutron Kinetics The neutron kinetics solver is employed to compute the steady-state volumetric fuel power distribution in the core. An eigenvalue approach is used for the purpose. Input:

- temperature field obtained in stage II, constant in time;
- total reactor power for flux normalization.

Output:

- effective multiplication factor;
- axial axial (radially integrated) flux distribution in the core, radial (axially integrated) flux distribution in the core;
- maximum, minimum and average flux in the core;
- maximum, minimum and average volumetric fuel power.

IV Thermal-mechanics The thermal-mechanics solver is employed to compute the steady-state axial and radial core expansion (i.e. displacement). Input:

- temperature field obtained in stage II, constant in time.

Output:

- maximum and minimum core radial displacement;;
- maximum and minimum fuel axial expansion.

Coupled Multi-physics Simulations

V Coupled Thermal-hydraulics The interplay between the temperature, velocity and pressure fields is assessed in this stage. Input:

- volumetric fuel power field obtained in stage III, constant in time;
- uniform pump momentum source, constant in time.

The output quantities consists in those outlined in stages I and II.

VI Coupled Thermal-hydraulics: Buoyancy The interplay between the temperature, velocity and pressure fields is re-assessed in this stage, with particular regard to buoyancy effects which arise from the prescribed volumetric fuel power density. Input:

- volumetric fuel power field obtained in stage III scaled down in magnitude by a factor 20, constant in time.

The volumetric fuel power field is scaled down as natural circulation cannot provide the same cooling power as a forced convection situation. The output quantities consists in those outlined in stages I and II.

VII Coupled Energy and Neutron Kinetics The interplay between the temperature and the volumetric fuel power fields is assessed in this stage. It is recalled that cross sections are parametrised, among others, with respect to temperature and coolant density (fuel density reduction, connected to the axial fuel thermal expansion is captured by a parametrisation with respect to the fuel displacement field, not highlighted in this stage). Input:

- velocity field obtained in Stage V;
- pressure field obtained in Stage V;
- null displacement field, constant in time.

The output quantities consists in those outlined in stages II and III.

VIII Thermal-hydraulics and Neutron Kinetics Stemming from step VI, the effect of the interplay between the velocity and temperature fields on neutron kinetics is assessed. Input:

- null displacement field, constant in time.

The output quantities consists in those outlined in stages I, II and III.

IX Energy, Neutron Kinetics and Thermal-mechanics Stemming from step VI, the effect of the interplay between the temperature, displacement and volumetric fuel power fields is assessed. Input:

- velocity field obtained in stage VIII.
- pressure field obtained in Stage VIII;

The output quantities consists in those outlined in stages I, II and III.

X Fully Coupled Multi-physics Simulation A fully coupled thermal-hydraulics, neutron kinetics and thermal-mechanics calculations is performed at steady state. Input:

- uniform pump momentum source, constant in time.

The output quantities consists in those outlined in all of the single-physics stages.

RESULTS

In the current section, results obtained by the application of the methodology to the GeN-Foam code are reported. Some remarks have to be made regarding some further simplifying assumptions that characterise this preliminary work. In the first place, the sodium flow is modelled as incompressible. In the second place, fluid density dependence on temperature is captured only via a Boussinesq buoyancy model. This means that changes in fluid density due to temperature are only reflected in the gravity momentum source term. In the third place, a simplified one-energy-group flux modelling was employed. The benchmark data regarding the ESFR concept taken as reference, system geometry, material and fluid properties, cross sections and cross section interpolation data are reported in the appendices. In Tab. 1, the results obtained by the various simulation stages discussed in the previous sections are presented. The axial and radial profiles of the various quantities in the core regions are not obtained in this preliminary stage of the work.

[PUT PICTURE OF T, U DISTRIB IN SYSTEM FOR STAGE X, FOR CLARITY]

	I	II	III	IV	V	VI	VII	VIII	IX	X
$u_{max} [\text{m s}^{-1}]$	3.577	—	—	—	—	—	—	—	—	—
$u_{min} [\text{m s}^{-1}]$	3.569	—	—	—	—	—	—	—	—	—
$\bar{u} [\text{m s}^{-1}]$	3.574	—	—	—	—	—	—	—	—	—
$\dot{M} [\text{kg s}^{-1}]$	18738	—	—	—	—	—	—	—	—	—
$\Delta p [\text{bar}]$	0.541	—	—	—	—	—	—	—	—	—
$T_{f,max} [\text{K}]$	—	826.02	—	—	—	—	—	—	—	—
$T_{f,min} [\text{K}]$	—	696.35	—	—	—	—	—	—	—	—
$\bar{T}_f [\text{K}]$	—	762.12	—	—	—	—	—	—	—	—
$\rho_{max} [\text{kg m}^{-3}]$	—	853.42	—	—	—	—	—	—	—	—
$\rho_{min} [\text{kg m}^{-3}]$	—	823.31	—	—	—	—	—	—	—	—
$\bar{\rho} [\text{kg m}^{-3}]$	—	838.15	—	—	—	—	—	—	—	—
$k_{eff} []$	—	—	1.03906	—	—	—	—	—	—	—
$\Phi_{max} [\text{cm}^{-2} \text{s}^{-1}]$	—	—	3.317×10^{15}	—	—	—	—	—	—	—
$\Phi_{min} [\text{cm}^{-2} \text{s}^{-1}]$	—	—	9.257×10^{14}	—	—	—	—	—	—	—
$\bar{\Phi} [\text{cm}^{-2} \text{s}^{-1}]$	—	—	2.266×10^{15}	—	—	—	—	—	—	—
$P_{vol,max} [\text{MW m}^{-3}]$	—	—	701.021	—	—	—	—	—	—	—
$P_{vol,min} [\text{MW m}^{-3}]$	—	—	195.507	—	—	—	—	—	—	—
$\bar{P}_{vol} [\text{MW m}^{-3}]$	—	—	446.102	—	—	—	—	—	—	—
$d_{r,max} [\text{mm}]$	—	—	—	0.047	—	—	—	—	—	—
$d_{r,min} [\text{mm}]$	—	—	—	0.003	—	—	—	—	—	—
$\bar{d}_r [\text{mm}]$	—	—	—	0.021	—	—	—	—	—	—
$d_{a,max} [\text{mm}]$	—	—	—	6.973	—	—	—	—	—	—
$d_{a,min} [\text{mm}]$	—	—	—	1.246	—	—	—	—	—	—
$\bar{d}_a [\text{mm}]$	—	—	—	4.037	—	—	—	—	—	—

TABLE 1. Summary of characteristic quantities obtained in the methodology simulation stages.

CONCLUSIONS

A preliminary simplified benchmark for SFRs was developed in the broader context of the development of multi-physics tools within the ESR-SMART project. In light of the different physics that govern the behaviour of SFRs and their interplay, a benchmark methodology for a simplified 2-D axial-symmetric SFR geometry was devised. In the first place, single-physics were tested individually for prescribed input fields as inputs for steady-state scenarios and the simulation outcome is characterized against a number of selected quantities of interest. Then, the impact of the coupling between the physics was investigated

and the quantities of interest re-obtained. The proposed methodology is suitable for application to multi-physics code regression tests and code-to-code verification purposes. The extension of the methodology to transient scenarios is envisioned, as well as the application to other multi-physics codes of interest, such as the FAST system code developed at the Paul Scherrer Institut.

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Appendix A: ESFR Data

The physical data

Appendix B: Benchmark Model Description

Appendix C: Benchmark Data

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